Design Plan Review — Resolutions

Generated 2026-04-21 · Scope: `pos-discount.md` · `call-center-design.md` · `bulk-pos-design-doc.md`

> **Summary.** 41 findings across three design docs. Each row below lists the finding, its severity, and the disposition folded back into the corresponding plan. Items marked DECIDE still need a tech-lead/finance call before the plan is implementation-ready. Items marked FIXED were already addressed in code during Part A.3.

Severity: HIGH correctness/security MED spec ambiguity LOW polish/hygiene **Status:** FIXED SPEC-UPDATED DEFER DECIDE

B.1 — pos-discount.md

B.1.1 Math / rounding

ID	Sev	Finding	Resolution	Status
1.1.a	HIGH	Rounding rule inconsistent between backend total, POS totals, and receipt. Line-rounded values can diverge from backend `total` by ±1 EGP.	Lock the rule: carry full precision through all intermediate computations; round only (a) persisted `discountAmount` and (b) final `total`. Receipt renders the backend-computed `total` verbatim; tax line shows the unrounded value rounded for display only.	SPEC-UPDATED
1.1.b	HIGH	§4.4 refund formula is mislabeled. "Proportional refund of the discount" does not match the code.	Rewrite §4.4: `proportional_discount_share = round1(lineSubtotal * discount_amount / original_subtotal)`; `refundAmount = lineSubtotal − proportional_discount_share`. Algebraically identical; matches the spec wording.	SPEC-UPDATED
1.1.c	MED	§4.4 references `original_subtotal`; no such column defined. Existing `sales.subtotal` semantics unclear.	Confirmed: `sales.subtotal` is stored pre-discount. Spec now states this explicitly; refund formula uses `sales.subtotal` as `original_subtotal`.	SPEC-UPDATED

ID	Sev	Finding	Resolution	Status
1.1.d	HIGH	Tax computed on *post-discount* subtotal changes tax_amount on the sale. May break statutory VAT treatment.	Needs a finance/legal call. Plan now flags this as an open decision; default keeps current "tax on post-discount" behavior until finance weighs in. If the answer changes, formula moves to `tax = subtotal * taxRate` (pre-discount) and receipt shows the discount as a post-tax concession.	DECIDE

B.1.2 Schema

ID	Sev	Finding	Resolution	Status
1.2.a	HIGH	`discount_applied_by_user_id` FK uses `ON DELETE SET NULL` — loses audit traceability if the cashier row is ever deleted.	Changed to `ON DELETE RESTRICT`. Users are soft-deleted elsewhere in the app; hard-delete of a user is already blocked by other FKs.	SPEC-UPDATED
1.2.b	MED	`discount_reason` is free-text VARCHAR(32). Typos corrupt grouped reports.	Add a `discount_reasons` lookup table (`code` PK, `label_ar`, `customer_label_ar`, `is_active`) seeded with the six codes in §3; FK from `sales.discount_reason`. Enables the customer-facing label addressed by 1.3.c.	SPEC-UPDATED
1.2.c	LOW	No backfill rule for historical call-center sales that already have discount data.	Historical rows get `discount_applied_by_user_id = NULL`. Documented; reports group NULL as "(Legacy)".	SPEC-UPDATED

B.1.3 State / UX

ID	Sev	Finding	Resolution	Status
1.3.a	MED	§7 shift-reopen / unfinished-sale discount-recovery is unresolved.	Decision: on cart draft restore, **clear** discount state (type, value, reason). Cashier must re-apply. Avoids stale / un-authorized residual discounts.	SPEC-UPDATED
1.3.b	LOW	When subtotal drops to 0, spec does not say whether the discount *inputs* (type/value/reason) are also cleared.	Clear them. Re-adding an item should not silently reapply a stale discount.	SPEC-UPDATED

ID	Sev	Finding	Resolution	Status
1.3.c	MED	Receipt shows internal reason codes like `MANAGER_COMP`. Customer-facing awkward.	Separate `label_ar` (internal) from `customer_label_ar` in the new `discount_reasons` table (1.2.b). Receipt renders the customer label only.	SPEC-UPDATED
1.3.d	LOW	Z-report discount summary does not say whether voided sales count.	Exclude voided sales from the Z-report discount totals (same rule as sales totals).	SPEC-UPDATED
1.3.e	LOW	No handling for `taxRate = 0`. Receipt prints "Tax (0%): 0".	Suppress the tax line when `taxRate == 0` or `tax == 0`.	SPEC-UPDATED

B.1.4 Audit / authz

ID	Sev	Finding	Resolution	Status
1.4.a	LOW	"Any cashier, no PIN/manager gate" is accepted but no abuse detection.	Add a Phase 2 report alert: flag sales where `discount_amount > 500 EGP` OR `discount % > 30%`. No pre-auth gate. Documented in plan's §4.3 Reports.	DEFER
1.4.b	LOW	`discount_applied_by_user_id` vs existing `audit_logs.actor_id` double-write risk.	Verify during implementation that both derive from `req.user.id` on the same request. No spec change.	DEFER

B.2 — call-center-design.md

B.2.1 Concurrency

ID	Sev	Finding	Resolution	Status
2.1.a	HIGH	§6.4 NEW → PREPARING is not atomic. Two cashiers can both claim; last write wins silently.	Spec now requires conditional update: `UPDATE orders SET status='PREPARING', claimedById=?, claimedAt=NOW() WHERE id=? AND status='NEW'`. Check `affectedRows == 1`; otherwise return 409 `ALREADY_CLAIMED`. Integration test required.	SPEC-UPDATED
2.1.b	HIGH	§6.2 shift-open drain fires for the opening user only. D6 says broadcast to all open-shift cashiers; the two rules contradict.	Drain broadcasts to *all* currently-open shifts at the branch via the EventBus. Clients dedup by `Order.id`. D6 is authoritative; §6.2 text updated to match.	SPEC-UPDATED
2.1.c	MED	§6.6 cancel transaction partial-failure semantics not explicit.	Cancel wraps `Inventory` credit + `Order.status='CANCELLED'` + `Sale` update in a single Prisma `$transaction`. Add an integration test that injects a failure after inventory credit to verify full rollback.	SPEC-UPDATED

B.2.2 Schema / migration

ID	Sev	Finding	Resolution	Status
2.2.a	MED	§5.2 backfill "create Order row for every existing Sale" can lock the table on large history.	Backfill runs in 5,000-row batches with a short sleep between batches; scheduled during the next maintenance window. Migration file documents the batching approach.	SPEC-UPDATED
2.2.b	MED	`Sale.branchId` and `Order.branchId` can drift.	Drop `Order.branchId`; always join through `Sale`. Indexes `[channel, status, branchId]` replaced by `[channel, status]` + join filter, with a compound index on `Sale(branchId, id)`. Schema section rewritten.	SPEC-UPDATED

ID	Sev	Finding	Resolution	Status
2.2.c	LOW	`Notification.payload` is `Json`, not indexable for "high-value order" filters.	Deferred. Current filters (userId, readAt) are enough for v1. Schema will extract `total` to a column when an indexed filter is first requested.	DEFER

B.2.3 SSE / real-time

ID	Sev	Finding	Resolution	Status
2.3.a	HIGH	§6.3 SSE auth uses JWT in URL query string — leaks through proxy logs, history, Referer.	Replaced with a single-use stream-ticket pattern: `POST /api/events/ticket` (authed) returns a 24-byte base64url token, 30s TTL, consumed on first use. `GET /api/events/stream?ticket=...` validates + deletes. Long-lived JWT never travels in a URL. Implemented in Part A.3.	FIXED
2.3.b	MED	Last-Event-ID replay against a 30-day-TTL Notification table: reconnect after > 30d silently misses events.	On reconnect, if `lastEventId < MIN(Notification.id)`, server emits a synthetic `replay-truncated` event. Client handles it by refetching state via REST. Added to §6.3 and §8 edge-cases.	SPEC-UPDATED
2.3.c	LOW	Heartbeat every 25s — Cloudflare Free closes idle at 100s.	25s is below that cap; no change. Noted: if deployment target changes to a different proxy, revisit interval. Documented inline in §6.3.	SPEC-UPDATED

B.2.4 Specification ambiguity

ID	Sev	Finding	Resolution	Status
2.4.a	MED	D9 shift attribution = "shift open at dispatch" — but the plan doesn't restrict who can dispatch.	Add authz rule to `PATCH /orders/:id/status` on the OUT_FOR_DELIVERY transition: actor must have an *open shift at the order's branch*. Otherwise return 403 `SHIFT_REQUIRED`. §6.4 updated.	SPEC-UPDATED
2.4.b	LOW	"Enable alerts" gate per-session unclear.	Persist the gate in `localStorage` (`alertsEnabled=true`). Cashier clicks once ever, not once per refresh.	SPEC-UPDATED
2.4.c	LOW	Notification TTL-purge index suboptimal at scale.	Accepted at v1 volume. Documented: at 2000 orders/day × 30d retention ≈ 60k rows, index scan is acceptable. Revisit if the TTL job runs > 10s.	SPEC-UPDATED

ID	Sev	Finding	Resolution	Status
2.4.d	LOW	Phase 1 includes a stub `RedisEventBus` file.	Removed from Phase 1 scope. Added to Phase 3 / post-v1 backlog when Redis is actually introduced.	SPEC-UPDATED

B.3 — bulk-pos-design-doc.md

B.3.1 Stored aggregates

ID	Sev	Finding	Resolution	Status
3.1.a	HIGH	`BulkOrder.paidAmount` / `balanceDue` are stored columns with no single write path — silent drift risk.	Pick option 2: every `BulkPayment` create/void/refund goes through a `bulkPaymentsService` that also recomputes `paidAmount` and `balanceDue` in the same transaction. Daily reconciliation cron asserts `paidAmount == SUM(non-voided payments)` and alerts on mismatch.	SPEC-UPDATED
3.1.b	MED	`getOpenBalance` clamps to 0 with `Math.max`, hiding over-payment.	Remove the clamp. Allow negative balances; surface as a "credit on file" badge on customer detail. Function now returns a signed number; callers (credit check, statement) handle both cases.	SPEC-UPDATED

B.3.2 Numbering

ID	Sev	Finding	Resolution	Status
3.2.a	MED	§4.4 issues `BULK-RCP-NNNN` but §3.7 has no `RECEIPT` sequence.	Add a `RECEIPT` key with prefix `BULK-RCP-` to `BulkNumberSequence` seed. §3.7 updated.	SPEC-UPDATED
3.2.b	MED	Atomic counter: "FOR UPDATE or retry loop" is ambiguous.	Pick `SELECT … FOR UPDATE` inside `prisma.$transaction` at the MySQL default REPEATABLE READ isolation level. Retry-loop variant dropped. `bulkNumbering.service.js` contract documented.	SPEC-UPDATED

ID	Sev	Finding	Resolution	Status
3.2.c	LOW	No sequence for PROFORMA, but `BulkDocument.documentNumber` is unique.	Proformas get their own sequence `PROFORMA` with prefix `BULK-PF-`. A `BulkDocument` row IS written when a proforma PDF is requested; subsequent requests for the same draft reuse the existing row. §3.7 + §4.3 updated.	SPEC-UPDATED

B.3.3 Credit limit

ID	Sev	Finding	Resolution	Status
3.3.a	HIGH	Confirm doesn't know the payment method yet (payments recorded separately), yet spec says "check if CREDIT".	At confirm, assume *worst case*: add the full `total` to open balance and reject if it exceeds limit. Every subsequent `POST /payments` recomputes; if a cash payment brings open balance below limit, no further action. Removes the ambiguity.	SPEC-UPDATED
3.3.b	LOW	`assertCreditLimitOk(tx, customer, projectedOrderTotal)` signature doesn't admit split payments.	Contract now: caller is responsible for computing `projectedOrderTotal` (worst-case unpaid portion). Service docstring updated. No signature change.	SPEC-UPDATED
3.3.c	MED	"Cash refund pays out but doesn't reduce open AR" — easy to misread.	Added three worked numeric examples to §4.5 (full refund of paid order, partial refund of partially-paid order, credit note against unpaid order). Each walks through `ordersSum − paymentsSum − refundsSum` to show the resulting open balance.	SPEC-UPDATED

B.3.4 Stock / fulfillment

ID	Sev	Finding	Resolution	Status
3.4.a	LOW	"One warehouse per order" not enforced at schema level.	Enforced by the `BulkOrder.warehouseId` scalar FK (one order = one warehouse). Confirmed no code path iterates multiple warehouses. Note added to §3.4.	SPEC-UPDATED

ID	Sev	Finding	Resolution	Status
3.4.b	HIGH	Confirm's stock check has no row lock — two concurrent confirms can oversell even though "block backorders" is the rule.	Spec now requires `SELECT … FOR UPDATE` on the affected `Inventory` rows inside the confirm transaction. Volume is tiny but correctness requires the lock. §4.4 updated.	SPEC-UPDATED
3.4.c	LOW	`BulkPayment.amount > 0` in prose only.	Added to the request validator (express-validator / zod, matching existing bulk controllers). §4.9 table row now reads `amount > 0 AND amount ≤ balanceDue`.	SPEC-UPDATED

B.3.5 Validation / API

ID	Sev	Finding	Resolution	Status
3.5.a	MED	`PUT /api/bulk/customers/:id/prices` — replace vs merge unclear.	PUT *replaces* the entire override list: any existing row not present in the body is deleted. Callers that want merge use new endpoint `PATCH .../prices`. §4.3 updated.	SPEC-UPDATED
3.5.b	LOW	No optimistic-concurrency token on `BulkOrder`.	Accepted at v1 volume (single-rep-per-customer convention, 5 orders/day). Documented explicitly. Revisit if > 2 reps per customer becomes common.	DEFER
3.5.c	LOW	Audit of Prisma Decimal may serialize oddly.	Unit test added to audit-capture middleware spec: assert that a `Decimal` field round-trips as a string (not `[object Object]`). Documented in §4.10.	SPEC-UPDATED

B.3.6 Phasing

ID	Sev	Finding	Resolution	Status
3.6.a	MED	Phase 1 defers tax-invoice but Phase 1 customers may have tax numbers.	Phase 1 blocks `POST /confirm` when `customer.taxNumber` is set, with error `TAX_INVOICE_NOT_AVAILABLE_IN_PHASE_1`. UI surfaces a blocker on the customer card. Removed at Phase 2.	SPEC-UPDATED
3.6.b	MED	Phase 1 has no payment recording — terminal status after confirm unclear.	Phase 1 confirm = `paymentMethod = CASH`, `paidAmount = total`, `status = PAID`, `paidAt = now()`, implicit cash receipt. Phase 2 replaces with full payment modal; existing PAID orders remain PAID.	SPEC-UPDATED

B.3.7 Reports

ID	Sev	Finding	Resolution	Status
3.7.a	MED	Aging bucket cutoffs are an open question.	Set to conventional **0–30 / 31–60 / 61–90 / 90+**. Configurable later via a `SystemConfiguration` entry if finance asks.	SPEC-UPDATED
3.7.b	LOW	No "discounts given" report — correct per §2.1, just confirm it sticks.	Confirmed: no discount in bulk. If added later, `BulkOrder` gets explicit discount columns mirroring the POS discount schema.	SPEC-UPDATED

Counts

Doc	Total	High	Med	Low	Fixed	Spec-updated	Defer	Decide
pos-discount.md	13	3	4	6	0	10	2	1
call-center-design.md	12	3	4	5	1	10	1	0
bulk-pos-design-doc.md	16	3	6	7	0	15	1	0
Total	**41**	**9**	**14**	**18**	**1**	**35**	**4**	**1**

What needs your call before implementation

- **1.1.d (HIGH)** — VAT: tax on gross (pre-discount) vs net (post-discount). Finance/legal decision. Default kept as post-discount until confirmed.

Source of truth: `plans/fix-consol-and-plan-review.md` (Part B). Each plan now carries a "Review Addenda (2026-04-21)" section at the end with these resolutions folded in. Open the corresponding plan to see the inline spec updates.